

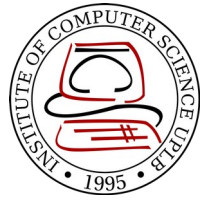
# *CMSC 137*

## *Data Communications and Networking*

Lecture Module  
21 - Routing



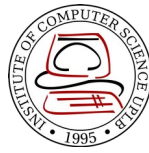
University of the Philippines  
LOS BAÑOS



# Module 21: Routing

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Lecturer

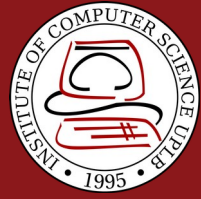
# Introduction



The goal of the network layer is to deliver a datagram from its source to its destination or destinations

Terms:

- **Forwarding** – the way a packet is delivered to the next station
- **Routing** – process of selecting a path for traffic in a network or across multiple networks
  - **Unicast Routing** – if a datagram is destined for only one destination
  - **Multicast Routing** – if a datagram is destined for several destinations

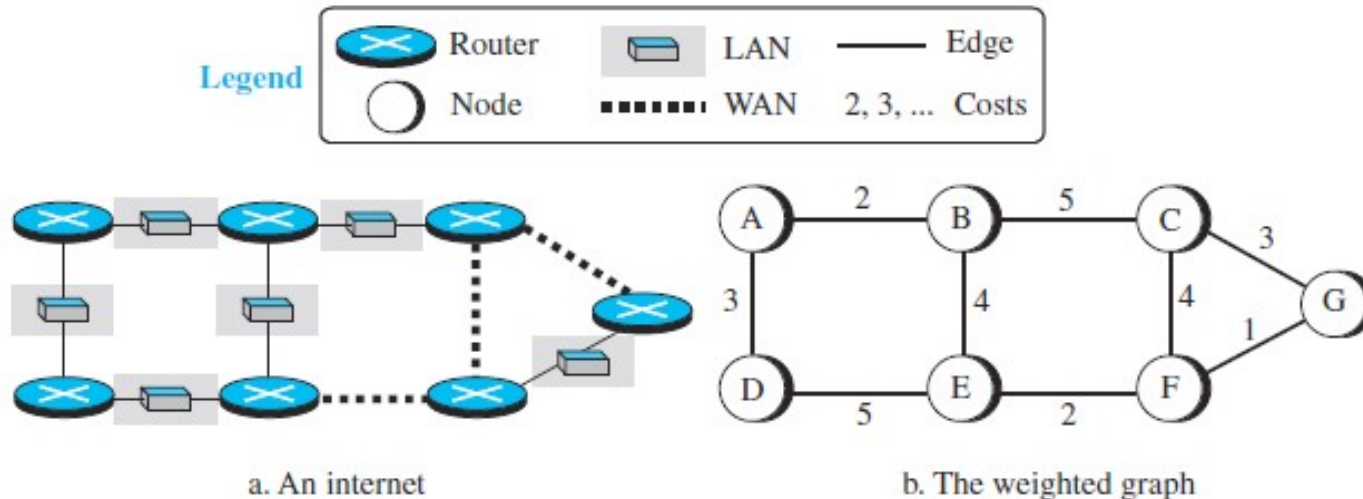


# Unicast Routing

# Unicast Routing



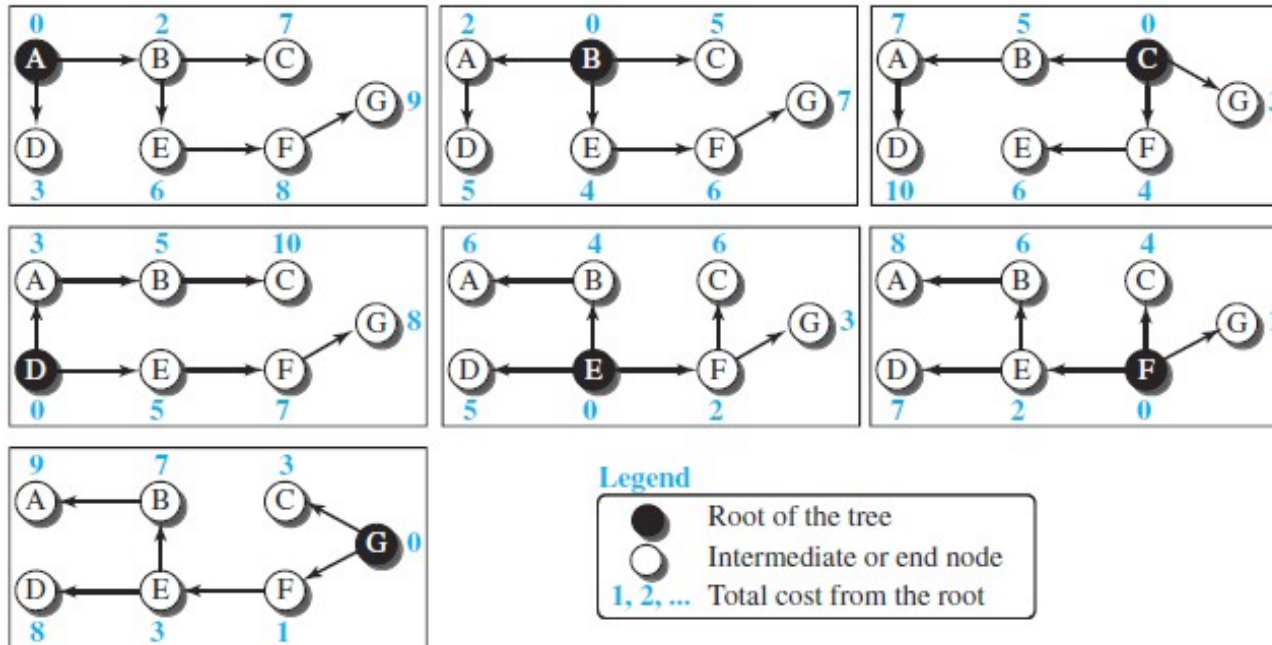
- A packet is routed, hop by hop, from its source to its destination by the help of forwarding tables
- To find the best route, an internet can be modeled as a graph



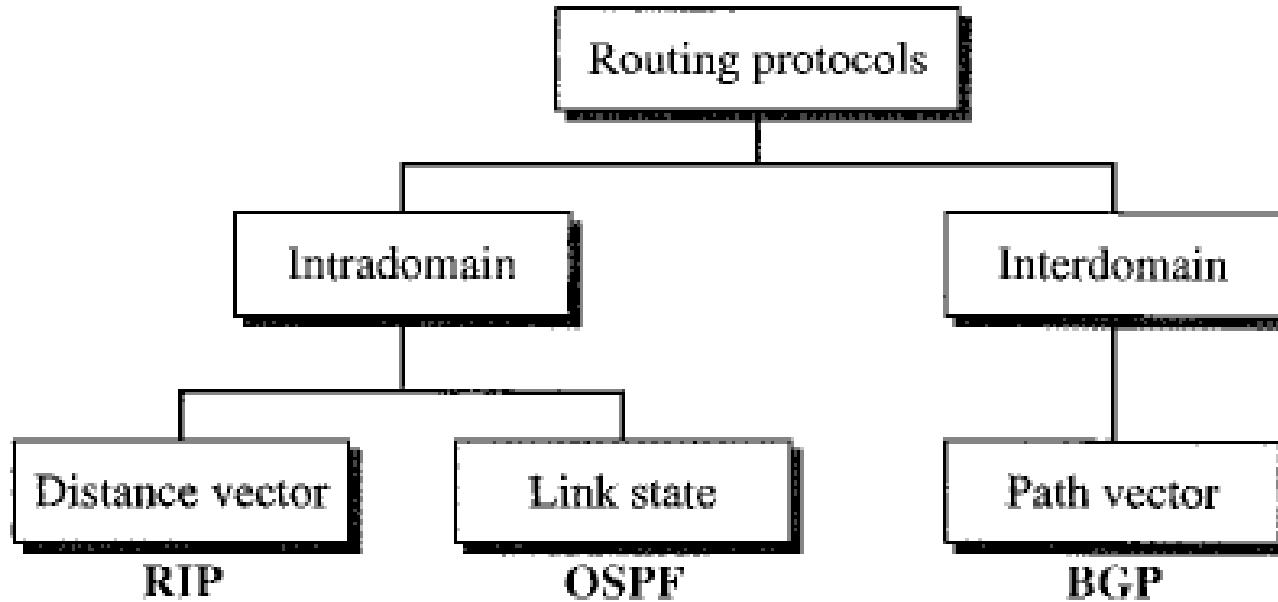
# Least Cost Routing



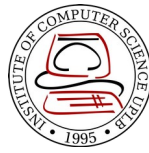
The source router chooses a route to the destination router in such a way that the total cost for the route is the least cost among all possible routes



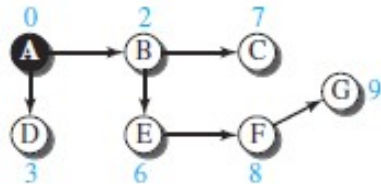
# Routing Algorithm



# Distance-Vector Routing



- Each node maintains a vector (table) of minimum distances to every node
- The table at each node also guides the packets to the desired node by showing the next stop in the route

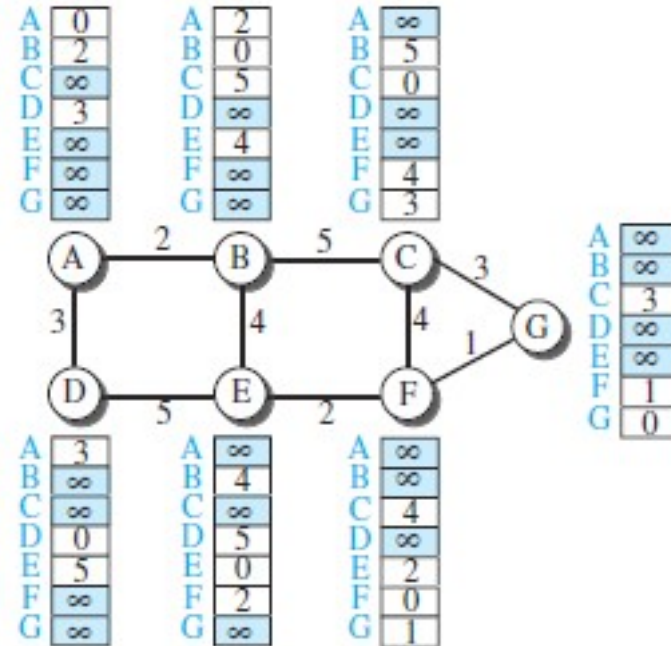


a. Tree for node A

A

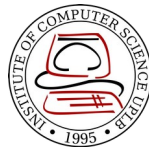
|   |   |
|---|---|
| A | 0 |
| B | 2 |
| C | 7 |
| D | 3 |
| E | 6 |
| F | 8 |
| G | 9 |

b. Distance vector for node A

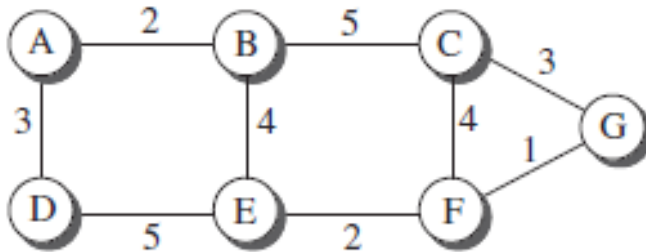




# Link-State Routing



- This method uses the term **link-state** to define the characteristic of a link (edge) that represents a network in the internet
- The collection of states for all links is called the **link-state database** (LSDB)
- To create a least-cost tree, each node needs a complete map of the network (using Dijkstra's Algorithm)

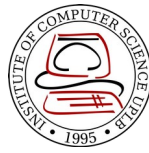


a. The weighted graph

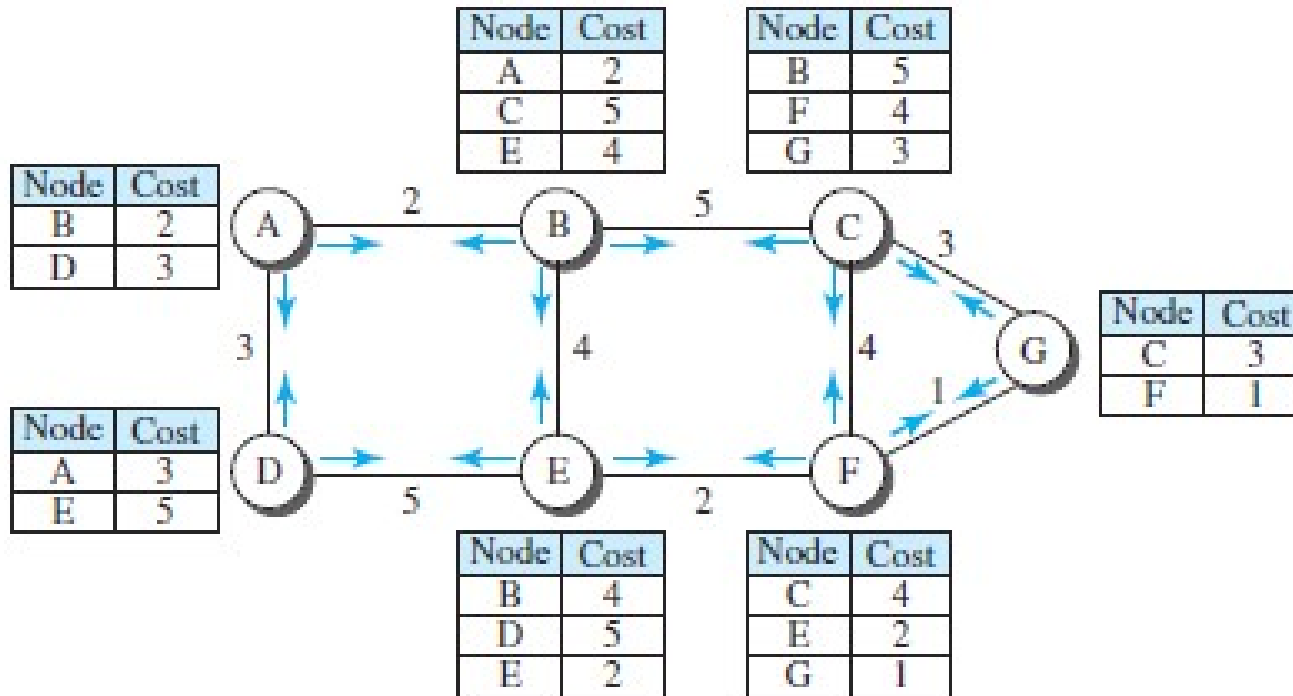
|   | A        | B        | C        | D        | E        | F        | G        |
|---|----------|----------|----------|----------|----------|----------|----------|
| A | 0        | 2        | $\infty$ | 3        | $\infty$ | $\infty$ | $\infty$ |
| B | 2        | 0        | 5        | $\infty$ | 4        | $\infty$ | $\infty$ |
| C | $\infty$ | 5        | 0        | $\infty$ | $\infty$ | 4        | 3        |
| D | 3        | $\infty$ | $\infty$ | 0        | 5        | $\infty$ | $\infty$ |
| E | $\infty$ | 4        | $\infty$ | 5        | 0        | 2        | $\infty$ |
| F | $\infty$ | $\infty$ | 4        | $\infty$ | 2        | 0        | 1        |
| G | $\infty$ | $\infty$ | 3        | $\infty$ | $\infty$ | 1        | 0        |

b. Link state database

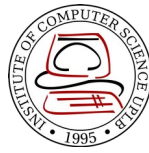
# Link-State Routing



LSPs created and sent out by each node to build LSDB

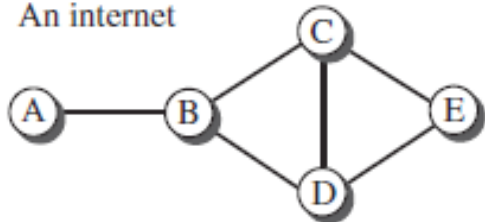


# Path-Vector Routing

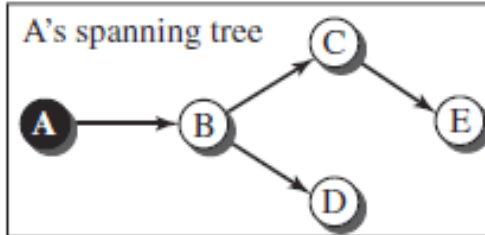


- The best path is determined by the source using the policy it imposes on the route, i.e. the source controls the path
- The path is also determined by the **best spanning tree**. It is not the least-cost tree; it is the tree determined by the source when it imposes its own policy

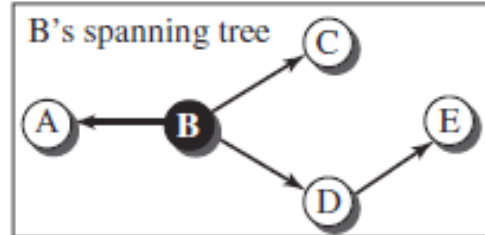
An internet



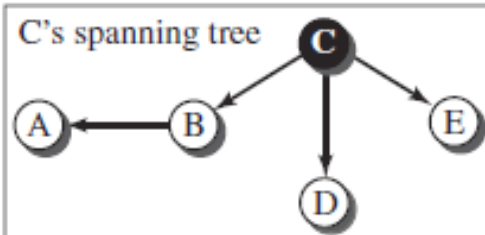
A's spanning tree



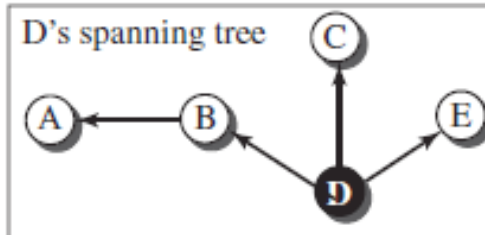
B's spanning tree



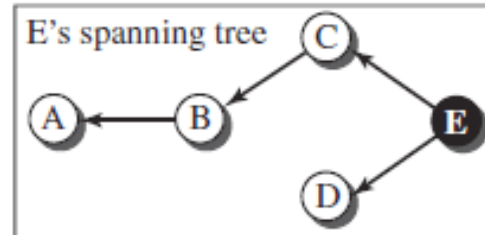
C's spanning tree



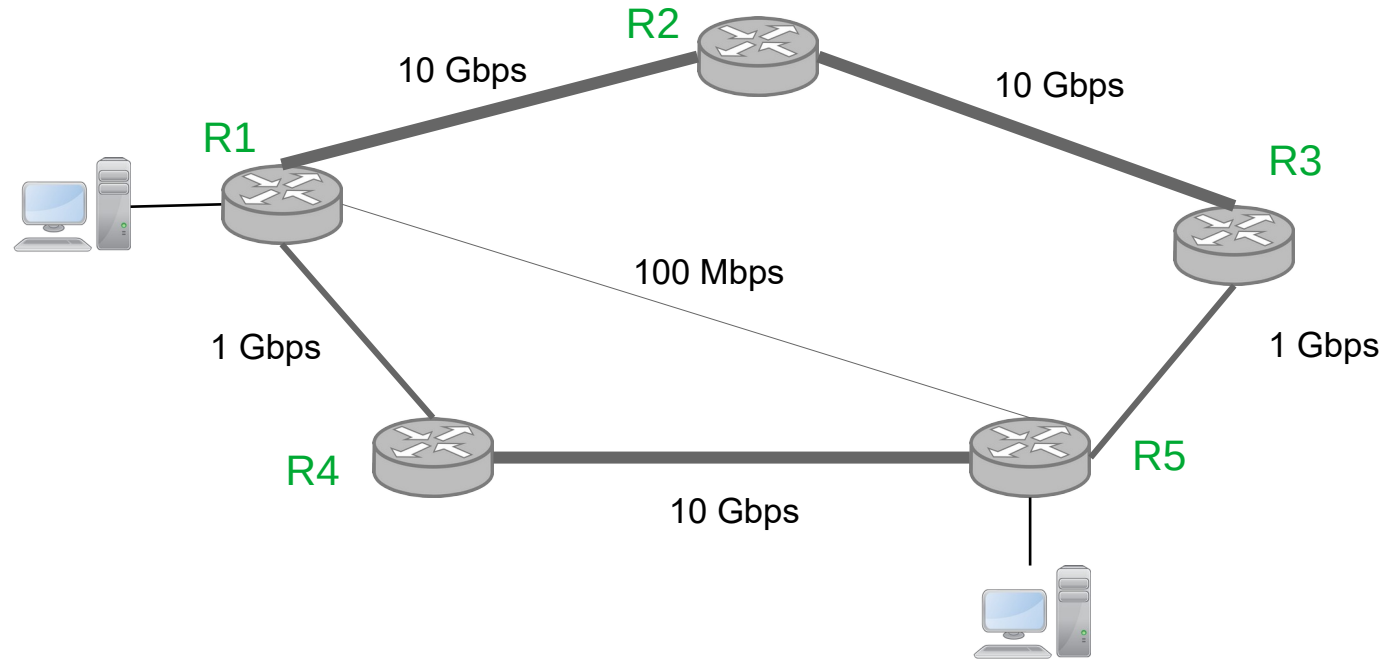
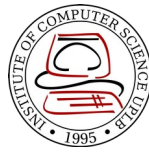
D's spanning tree



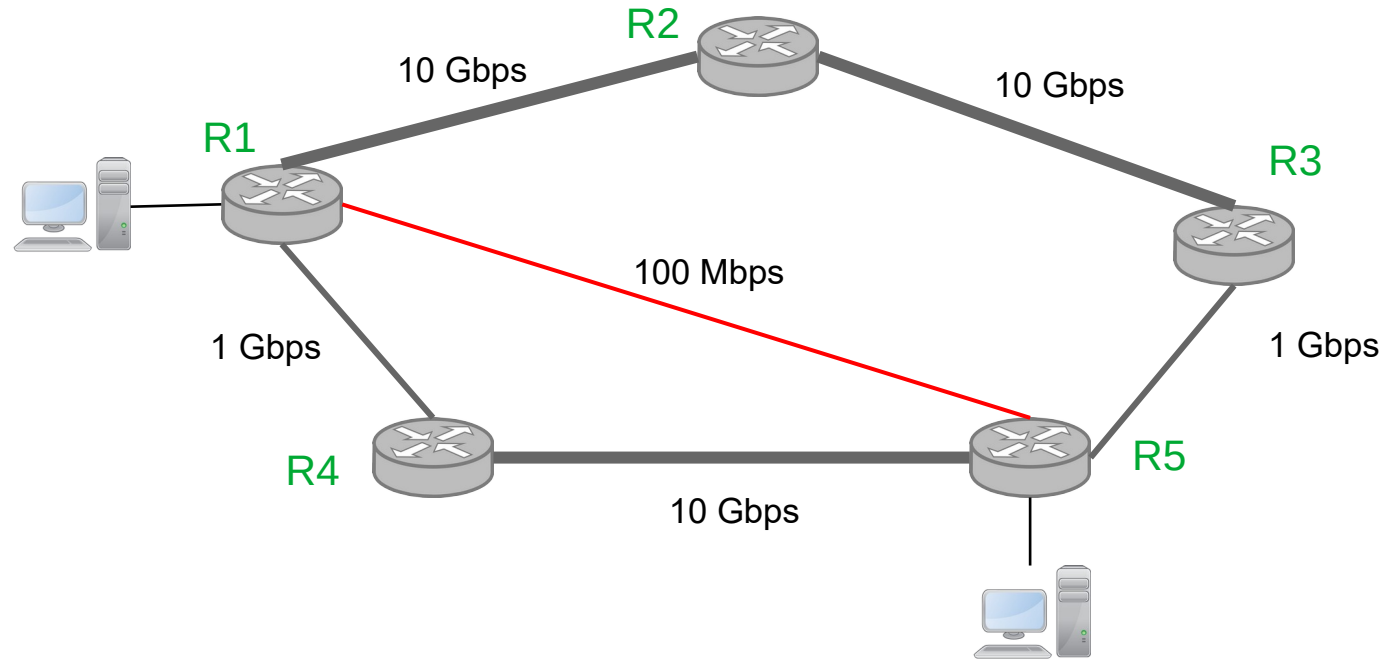
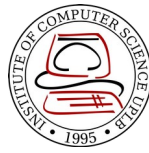
E's spanning tree



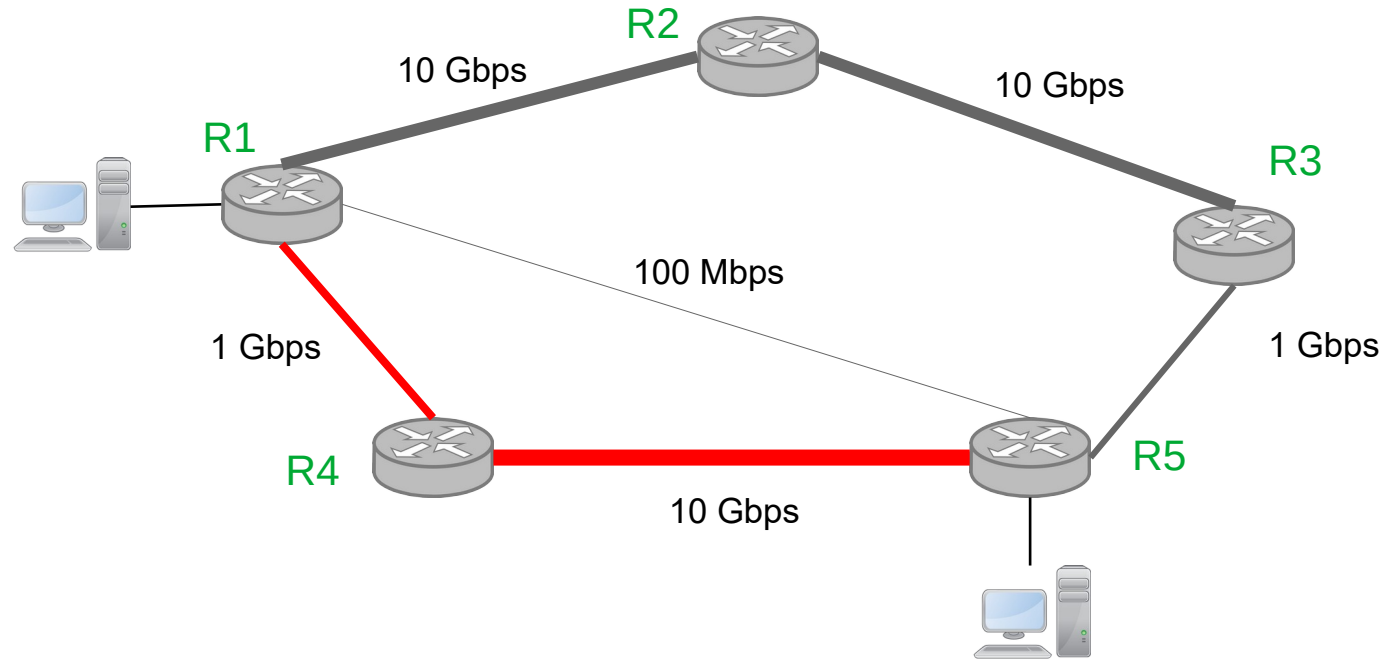
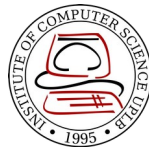
# Example



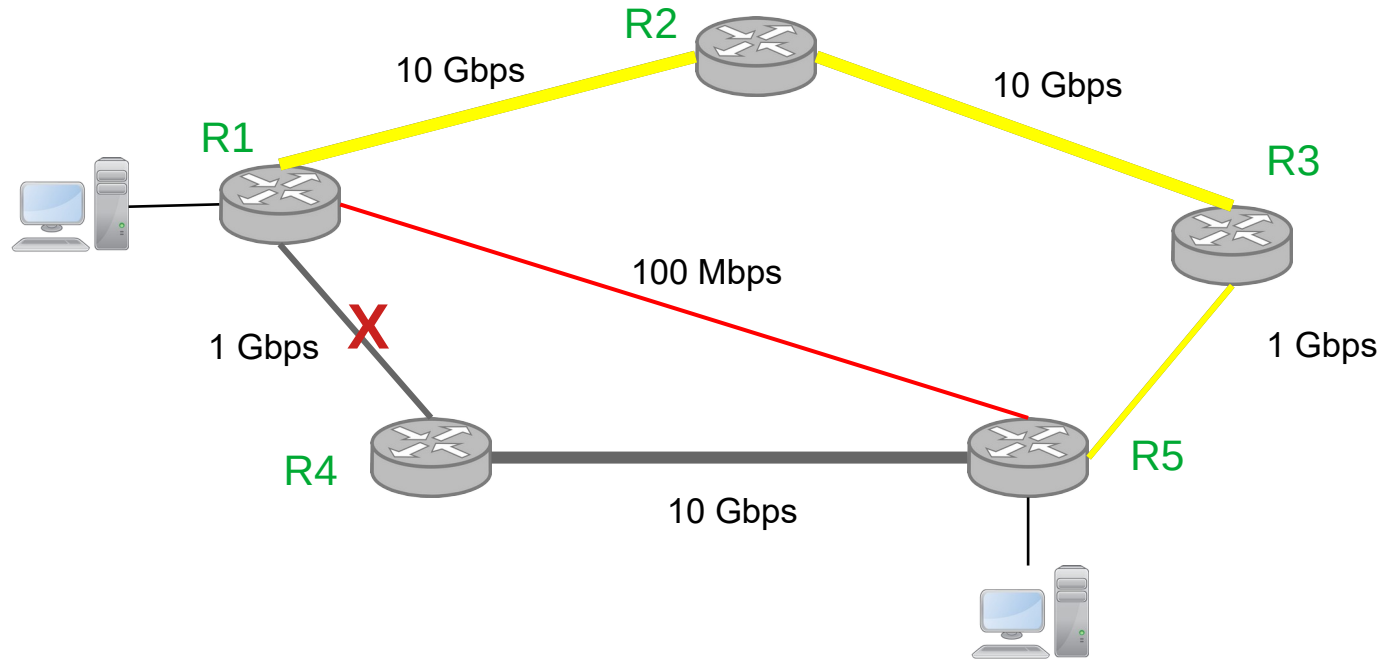
# Distance-Vector Routing



# Link-State Routing



# Path-Vector Routing



Policy  
R1 and R4 not allowed

# References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.